

Name: _____

Date: _____



SOLUTIONS

Year 8 Mathematics

Test 4, 2015

Topics – Linear Relationships

55
= ____ %

Total Time: **60** minutesTotal Reading: **5** minutesTotal Working: **55** minutes

SECTION 1: CALCULATOR FREE

Time:	36 minutes	Marks for Section 1:	33
Reading:	3 minutes	Equipment Allowed:	$\frac{1}{2}$ page notes (A4 one side)
Working:	33 minutes		

1. [6 marks: 3, 1, 2]

Complete the sentences from the list of words below:

Cartesian linear graph gradient
x-axis variable y intercept y- axis

- a) The Cartesian plane has two axes. The Horizontal axis is called the x-axis and the vertical axis is called the y-axis.
- b) A rule that produces a straight line is called a linear function.
- c) The gradient of a line is also called the slope and the y-intercept is where the line touches the y axis.

2. [4 marks: 2, 2]

Complete the tables below:

a)

x	1	2	3	4	5	6
y	5	8	11	14	17	20

b)

x	1	2	3	4	5	6
y	13	8	3	-2	-7	-12

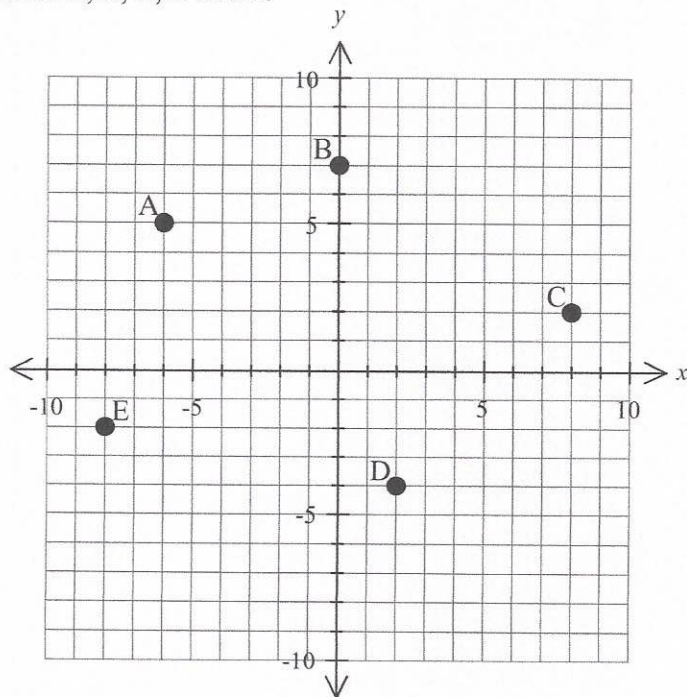
(-1 each error)

✓✓

✓✓

3. [5 marks: 1, 1, 1, 1, 1]

Write down the coordinates for A, B, C, D and E.



A (-6, 5) ✓

B (0, 7) ✓

C (8, 2) ✓

D (2, -4) ✓

E (-8, -2) ✓

4. [4 marks: 1, 1, 1, 1]

Plot the following coordinates on the graph below.

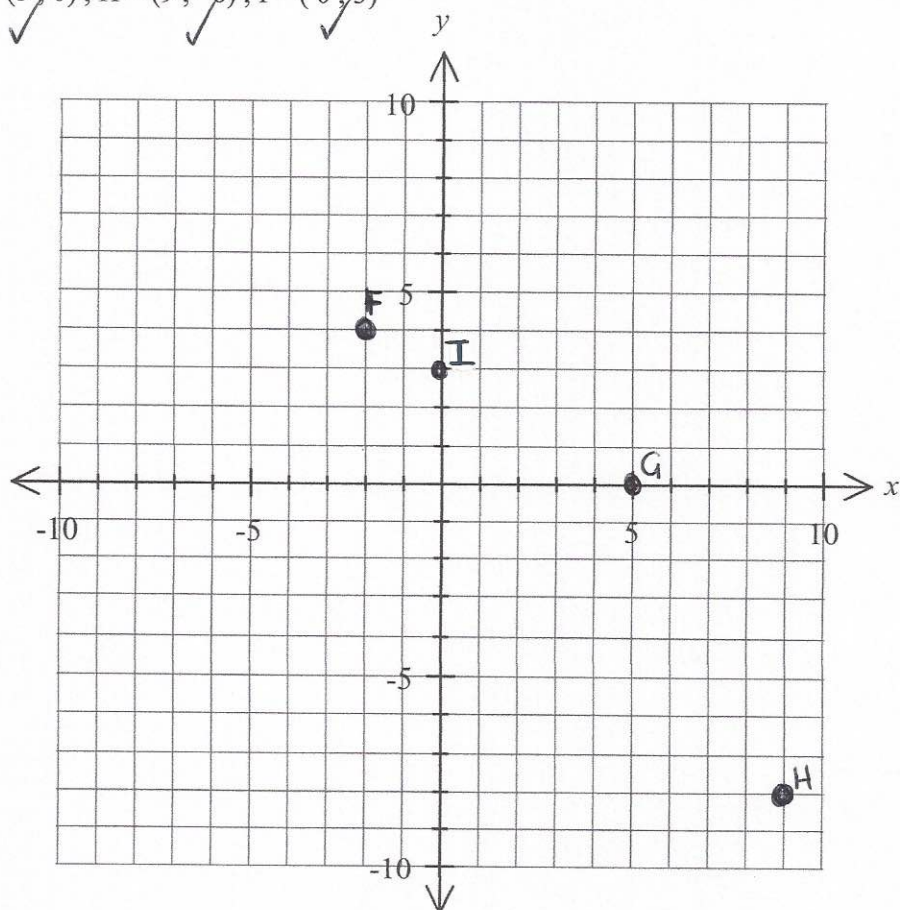
F = (-2, 4), G = (5, 0), H = (9, -8), I = (0, 3)

✓

✓

✓

✓



5. [4 marks: 2, 2]

Complete the following tables:

($-\frac{1}{2}$ each error)

a) $y = 4x + 2$

x	0	1	2	3	4	10
y	2	6	10	14	18	42

✓✓

b) $y = -2x - 5$

x	0	1	2	3	4	10
y	-5	-7	-9	-11	-13	-25

✓✓

6. [4 marks: 2, 2]

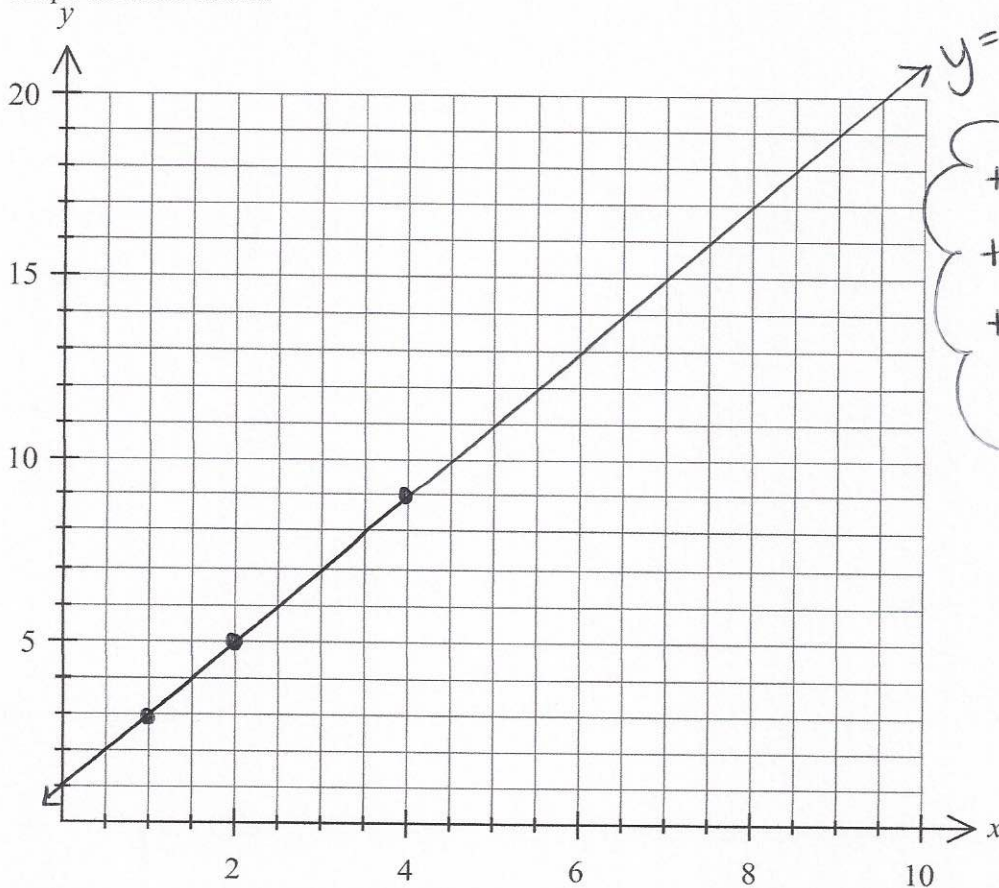
x	1	2	3	4	5	6
y	3	5	7	9	11	13

+2 +2 +2 +2 +2

a) From the table above write a rule in the form of $y = \dots\dots\dots$

$y = 2x + 1$

b) Graph the table above.



+ $\frac{1}{2}$ arrows
+ $\frac{1}{2}$ joined by line
+ 1 points plotted correctly.

7. [6 marks: 2, 2, 2]

For each of the following tables, determine if they show a linear relationship. Explain why.

a)

x	1	2	3	4	5	6
y	10	14	18	22	26	30

Linear relationship: YES/NO

Explanation:

constant first difference pattern ✓

b)

x	1	2	3	4	5	6
y	1	3	9	27	81	243

Linear relationship: YES/NO

Explanation:

not a constant first difference pattern ✓

c)

x	1	2	3	4	5	6
y	100	89	78	67	56	45

Linear relationship: YES/NO

Explanation:

constant first difference pattern ✓

~ END OF TEST SECTION 1 ~

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SECTION 2: CALCULATOR ASSUMED

Time:	24 minutes	Marks for Section 2:	22
Reading:	2 minutes	Equipment Allowed:	<i>½ page notes (A4 one side), Scientific calculator</i>
Working:	22 minutes		

8. [5 marks: 1, 1, 1, 2]

Calculate the gradient and the y-intercept for the following:

	Equation	Gradient	y - intercept
a)	$y = 5x - 2$	+5 $\frac{1}{2}$	-2 $\frac{1}{2}$
b)	$y = -6x$	-6 $\frac{1}{2}$	0 $\frac{1}{2}$
c)	$y = 10 - 2x$	-2 $\frac{1}{2}$	+10 $\frac{1}{2}$
d)	$2y + x = 4$	$-\frac{1}{2}$ ✓	+2 ✓

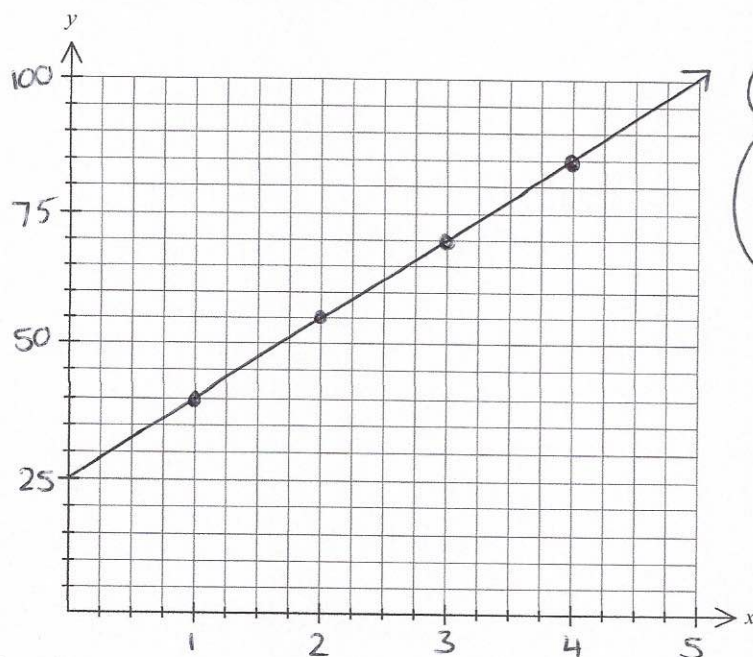
9. [5 marks: 2, 2, 1]

A plumber charges \$25 an hour plus a set call out charge of \$15.

- a) Create a rule to show the plumbers total cost (c), with hours (h).

$$C = 25 + 15h$$

- b) Plot your rule on the graph paper below.



+½ arrows
 +½ joined by line
 +1 points plotted correctly

- c) How much would it cost to have the plumber for 12 hours?

$$\$205 \quad \checkmark$$

10. [8 marks: 2, 2, 2, 2]

Draw the following equations on the graph below:

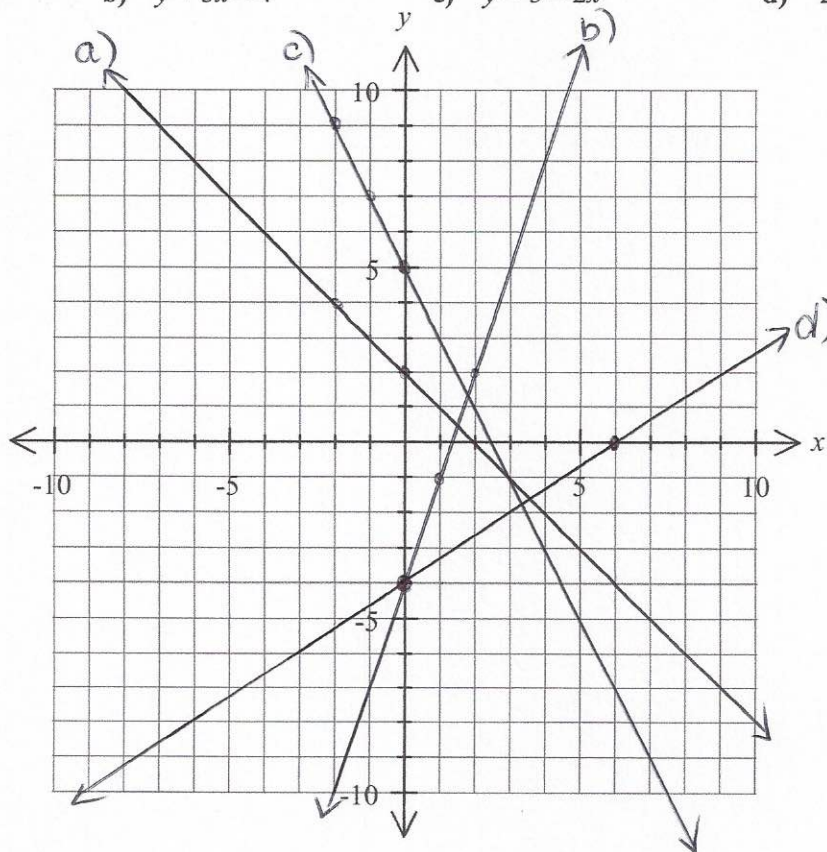
a) $y = -x + 2$

b) $y = 3x - 4$

c) $y = 5 - 2x$

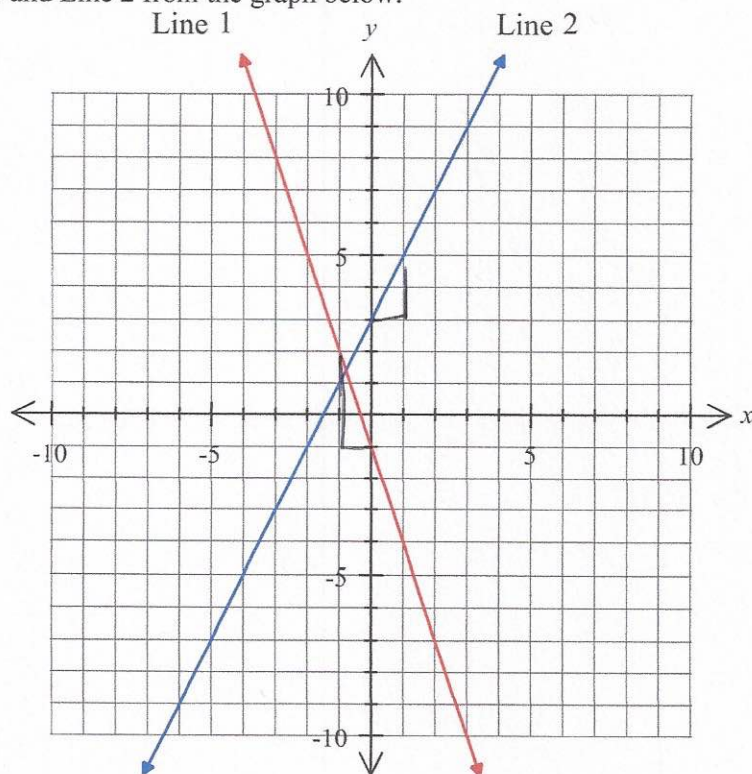
d) $2x - 3y = 12$

for each:
 + $\frac{1}{2}$ arrows
 + $\frac{1}{2}$ label
 + $\frac{1}{2}$ gradient
 + $\frac{1}{2}$ y int.



11. [4 marks: 2, 2]

Find the rule for Line 1 and Line 2 from the graph below.



Line 1 $y = -3x + 1$

Line 2 $y = 2x + 3$

~ END OF TEST SECTION 2 ~